TEXORA · LECTURE SHEET

Management of Textile and Apparel Technology

How a textile firm chooses, acquires, absorbs and eventually replaces its technology: life cycles of technologies and products, automation and digital tools, evaluation and the cost of standing still.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 4 · LEVEL 3 TERM 1

1. Technology and its management

Technology is the practical application of knowledge to convert inputs into outputs: not only machines, but the complete bundle of **hardware** (equipment, instruments), **software** (programmes, recipes, data), **know-how** (skills, routines, operating discipline) and **support systems** (maintenance, spares, training, information flow) that together make a process work at its intended quality and cost. A modern air-jet loom delivered without trained technicians, weaving preparation know-how and a spares pipeline is not yet a technology in the factory — it is expensive furniture. This bundle view explains why two factories buying the identical machine earn different returns from it.

Technology management is the planning, organising and control of that bundle across its whole life: identifying which technologies the firm's strategy needs, evaluating and acquiring them, implementing and absorbing them into production, exploiting them profitably, protecting the know-how they create, and retiring them at the right time. In textiles the decisions are large and long-lived — a dyeing range, a spinning plant or an automatic cutting room commits capital for a decade or more, fixes the firm's cost structure, quality ceiling and product range, and cannot be reversed cheaply. Technology management is therefore a top-management subject, not a purchase-department errand.

2. The technology life cycle

Technologies, like products, pass through a recognisable life cycle. Understanding which stage a technology occupies tells a manager whether to invest, exploit, defend or exit.

- **Emergence / development.** The technology exists in laboratories and pilot lines; performance is uncertain, cost is high, standards are unsettled. Example in living memory: early digital textile printing — slow, costly, confined to samples and very short runs.

- **Growth.** Performance improves rapidly, prices fall, suppliers multiply and adoption spreads. Firms entering now gain capability while the technology is still a differentiator.
- **Maturity.** The technology is standard, reliable and widely available; competition shifts from having it to using it cheapest and best. Compact spinning and automatic cone winding sit here today: no longer a novelty, simply expected in a modern mill.
- **Decline / substitution.** A superior technology erodes the old one's ground; investment stops, spares and skills become scarce, and remaining users serve shrinking niches. Shuttle looms survive in speciality and traditional weaving long after air-jet and rapier looms took the volume market — a classic decline-stage pattern.

The trap is timing in both directions: buying at emergence pays pioneer's prices for pioneer's problems; buying deep into maturity is safe but buys no advantage; and sweating a decline-stage asset too long quietly raises cost per unit until the firm cannot meet market prices at any efficiency. The technology life cycle must also be distinguished from the **product life cycle** (Section 4): a mature technology can still make products anywhere in their own cycles.

3. Acquiring technology — sources and transfer

ROUTE	HOW IT WORKS	ADVANTAGES	RISKS / COSTS
Purchase of machinery and turnkey plants	Buy equipment, often with installation, commissioning and training from the maker	Fastest route to proven capability; supplier support	Capital-heavy; know-how may stay shallow if training is neglected
Licensing / technical collaboration	Pay for the right to use patented processes, brands or designs, often with technical assistance	Access without research cost; market credibility	Royalties; dependence; restrictions on markets or volumes
Joint venture / foreign partnership	Share ownership with a technology holder	Deep transfer, shared risk, buyer access	Shared control; complex agreements
In-house development (R&D)	Own laboratory and engineering effort adapts or creates technology	Owned know-how fitted exactly to the firm's products	Slow, uncertain, needs scarce research staff
Imitation / reverse adaptation	Study available products and processes and reproduce their principles legally	Cheap learning for follower firms	Patent limits; always one generation behind
Recruiting experienced technologists	Hire people who carry the know-how from previous employers	Immediate, practical knowledge	Knowledge leaves when they do, unless captured in systems

Whatever the route, **technology transfer** succeeds only through **absorption**: the receiving firm must train its people, adapt settings and recipes to local fibre, water, power and climate, build maintenance and spares capability, and document the process in its own operating procedures. The classic failure — seen in dye houses that bought computerised colour matching but kept shade decisions purely visual — is hardware installed and know-how never absorbed, so the old method runs beside the new machine until the machine is quietly abandoned. A transfer plan therefore budgets as seriously for training, trial production and documentation as for the machine itself, and judges the project complete only when local staff run it unaided at rated quality and speed.

4. The product life cycle (PLC) and technology decisions

The **product life cycle** describes a product's sales history in four stages — **introduction, growth, maturity, decline** — and each stage demands different technology and management choices. Fashion apparel shortens every stage: a garment style may live for one season, while the machines that make it must pay back over hundreds of styles.

STAGE	MARKET BEHAVIOUR	TECHNOLOGY / OPERATIONS EMPHASIS	TEXTILE EXAMPLE
Introduction	Low sales, high unit cost, buyers must be convinced	Flexibility over speed; sampling capability; small lots; rapid prototyping (CAD patterns, digital sample printing)	A new technical textile (e.g., a locally developed workwear fabric) offered to first buyers
Growth	Sales climb fast; competitors enter	Capacity expansion, quality consistency at volume, cost reduction begins; add machines, second sources	Basic knit T-shirts for a new export market scaling from trial orders to full programmes
Maturity	Sales peak and flatten; price competition intense	Efficiency technology dominates: automation, work study, energy and water saving; differentiation by finish, hand-feel, delivery reliability	Standard polo shirts and denim jeans — won on cost, quality and lead time, not novelty
Decline	Demand falls as substitutes or fashion move on	Harvest the product: no new investment, run existing machines, redeploy capacity to growth products	A fading fabric construction kept only for a shrinking set of loyal buyers

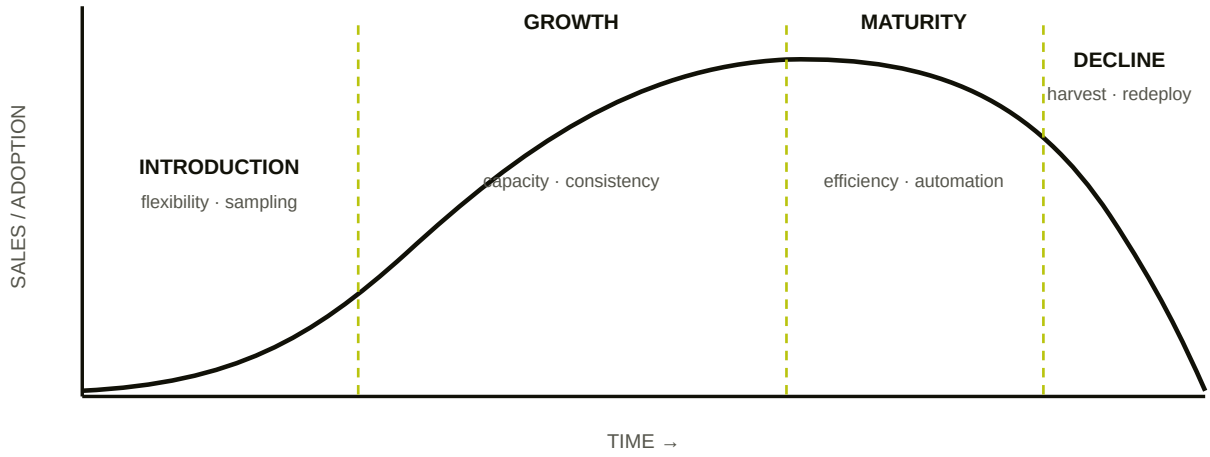


Figure 1 — The product life cycle (the technology life cycle follows the same S-curve). The curve's shape decides the technology strategy: flexible at introduction, capacity in growth, efficiency in maturity, no new investment in decline.

5. Innovation, automation and digital technology in textiles

Innovation is the successful commercial use of a new idea — distinct from invention, which only creates the idea. In this industry innovation is usually incremental and practical: a finish that gives the same softness with half the water, a cutting-room nesting programme that saves 1.5 % of fabric on every order, a dyeing recipe redesigned for right-first-time production. Such small gains, compounded across millions of pieces, are the industry's real profit engine — and they come from a managed system (suggestion schemes, trial budgets, pilot lines, tolerance for failed experiments) more than from lone genius.

Automation replaces manual effort and judgement with machine control in steps: mechanisation (power replaces muscle), single-machine automation (auto-doffing, auto-piecing, automatic cone winding), linked systems (transport and data flowing between processes), and computer-integrated production. In garments — where limp fabric resists full automation — progress concentrates around the sewing line: computerised pattern making and grading, CAD marker making, automatic spreading and CNC cutting, template sewing machines, unit production (hanger) systems that move garments between stations automatically, and real-time line data from tablet or RFID tracking.

Digital and Industry 4.0 tools now layer information over the physical factory: **CAD/CAM** in design and cutting; **computerised colour measurement** and recipe prediction in dyeing; **ERP (enterprise resource planning)** software joining orders, inventory, production and accounts in one database; machine sensors feeding downtime and efficiency data to dashboards; and data analysis that predicts maintenance before breakdown (predictive maintenance). The management lesson repeats Section 1: value comes from the bundle — data discipline, trained users and revised procedures — not from the

licence or the sensor alone. A factory whose floor data is entered carelessly will automate the production of confident, wrong numbers.

AREA	TRADITIONAL PRACTICE	TECHNOLOGY-MANAGED PRACTICE	GAIN SOUGHT
Pattern & cutting	Manual patterns, hand markers, hand cutting	CAD patterns and markers, auto-spreading, CNC cutting	Fabric saving, accuracy, faster style change
Dyeing	Visual shade judgement, manual dosing	Spectrophotometer, recipe software, automatic dosing and machine control	Right-first-time rate, reproducibility, chemical and water saving
Sewing	Bundle system, paper tickets, end-of-day counts	Template machines, hanger lines, real-time production tracking	Line balancing, early bottleneck warning, less WIP
Planning & control	Separate notebooks and spreadsheets per department	Integrated ERP with one order and inventory database	Single version of the truth; faster, evidence-based decisions
Maintenance	Repair after breakdown	Preventive schedules, condition sensors, spares planning	Higher machine availability, fewer shipment-threatening stops

6. Evaluating a technology choice

Before capital is committed, a proposed technology is tested against technical, economic, human and strategic questions. **Technically:** does it meet the required quality and product range, at what reliability, with what utilities (power, water, steam, compressed air) and effluent load? **Economically:** total cost of ownership — purchase, installation, training, spares, energy, maintenance — against savings in labour, material, time and quality failure, expressed through payback period and return on investment over the machine's life, not its first year. **Humanly:** can present staff be trained to run it, what jobs change, and how are displaced workers redeployed? **Strategically:** does it open the product markets the firm intends to serve in five years, satisfy buyers' compliance and traceability expectations, and keep options open (modular expansion) or lock the firm in? Pilot trials on the firm's own fibre and fabrics, reference visits to factories already running the technology, and written performance guarantees in the purchase contract are the standard protections. The cheapest machine that fails the firm's shade reproducibility is not cheap; the dearest machine whose capacity the order book cannot fill is not an investment but a monument.

> **OBSOLESCENCE — THE SILENT COST**

> Every technology choice also chooses a rate of obsolescence. Managing it means: keep machines earning through their economic (not merely physical) life, watch the substitution curve of the next technology, and time replacement so the firm sells or redeploys ageing capacity while it still has value — instead of discovering, in a price war, that its cost base belongs to a previous generation.

Glossary

Technology — the full bundle of hardware, software, know-how and support systems that makes a process work. **Technology management** — planning, acquiring, absorbing, exploiting and retiring technology in line with strategy. **Technology transfer** — movement of technology between firms or countries; completed only by absorption. **Absorption** — building local capability to run, maintain and improve a transferred technology unaided. **Product life cycle (PLC)** — introduction, growth, maturity and decline stages of a product's sales. **CAD/CAM** — computer-aided design / manufacturing. **ERP** — enterprise resource planning software integrating orders, stock, production and accounts. **Obsolescence** — loss of economic value as a superior technology replaces an older one. **Payback period** — time for a project's savings to repay its cost.

Key points

• Technology is a bundle — hardware, software, know-how, support — and returns depend on the whole bundle, never the machine alone.
• Technologies follow a life cycle (emergence, growth, maturity, decline); the stage dictates whether to pioneer, invest, exploit or exit.
• Acquisition routes — purchase, licence, joint venture, in-house development, imitation, hiring — all succeed only through planned absorption.
• The product life cycle demands different technology at each stage: flexibility, then capacity, then efficiency, then harvesting.
• Innovation in textiles is mostly incremental and systemic; automation and digital tools pay back only with data discipline and trained users.
• Evaluate every technology technically, economically (life-cycle cost), humanly and strategically before committing long-lived capital.

Practice

1. Place each technology in its life-cycle stage and justify: (a) shuttle looms for commodity sheeting; (b) digital textile printing; (c) compact spinning; (d) automatic cone winding. State the investment attitude each stage implies.

2. A mill buys a computerised colour-matching system, but six months later shade approval is still purely visual and the instrument is unused. Using the bundle definition of technology, explain what was bought, what was missing, and design the absorption plan that should have accompanied the purchase.
3. A basic polo-shirt programme is in maturity while a new performance-fabric line is in introduction. Contrast the technology priorities of the two programmes under the PLC framework.
4. Prepare the evaluation checklist a knit-dyeing firm should apply before buying a new low-liquor dyeing range: at least three questions under each of technical, economic, human and strategic headings.
5. "The cheapest machine is the one with the lowest invoice price." Attack or defend this statement using total cost of ownership.
6. Explain why garment sewing has automated more slowly than spinning or weaving, and identify where automation has succeeded in the garment factory and why there.

Answer key

1. (a) Decline — displaced by shuttleless looms for volume work; invest nothing beyond maintenance, harvest remaining niche value. (b) Growth (after emergence) — performance improving, adoption spreading; early investment still differentiates. (c) Maturity moving from growth — standard expectation in quality spinning; invest for competitiveness, judge on efficiency of use. (d) Maturity — universal in modern mills; buy on reliability and cost, not novelty. Credit correct stage logic with justification.
2. Bought: hardware and software only. Missing: know-how (trained colourists trusted to use data), revised procedures (instrumental pass/fail criteria agreed with buyers), and support (calibration, standard illuminants, maintenance). Absorption plan: train and certify lab staff before installation; run parallel visual/instrumental approval for a trial period with comparison records; agree numerical tolerance with key buyers; make the lab manager accountable for usage rate and right-first-time statistics; review at 3 and 6 months.
3. Maturity (polo): efficiency technology — automation of cutting and materials handling, work-study and SMV reduction, energy/water saving, defect prevention; no capacity expansion without orders. Introduction (performance fabric): flexibility technology — sampling and pilot capability, small-lot dyeing, rapid testing and prototyping, close buyer development; accept high unit cost while learning. The contrast — cost-down versus learn-fast — is the required point.
4. Technical: liquor ratio and reproducibility on the firm's fabric range; utility consumption per kg; effluent load and compatibility with the ETP; reliability and local service. Economic: installed cost including utilities and building work; savings in water, energy, chemicals and reprocessing versus present machines; payback at realistic utilisation; spares pricing. Human: training provided and in

what language/depth; operator and maintenance skill gaps; staffing change and redeployment plan. Strategic: fit with the five-year product and buyer plan; compliance/traceability features (data logging); expandability; supplier's installed base and survival prospects.

5. Attack: invoice price is one term in total cost of ownership. Energy and water per kg, reprocessing rate from poor reproducibility, downtime and spares prices, training burden and resale value routinely outweigh a small invoice saving over a 10–15-year life. A machine 10 % cheaper that reprocesses 3 % more dye lots is the expensive machine. (A limited defence — identical performance and support — merely proves the rule by assuming the other costs equal.)
6. Spinning and weaving handle fibres and yarns in continuous, mechanically regular forms that machines grip and control easily; a limp, stretchy fabric panel changes shape in the hand, so presenting, aligning and sewing it long required human dexterity and judgement. Automation therefore succeeded first where material can be made rigid or the task standardised: CAD/cutting (flat, measurable), template sewing (fabric clamped in a frame), hanger transport (movement, not manipulation), and data systems — and spreads operation-by-operation as clamping and sensing improve.
